

WE HAVE REALIZED THAT WE CAN'T COMPUTE THE TOTAL FIELD USING THE WAVE EQUATION SOLUTION, BECAUSE $k(\underline{r})$ IS NOT A CONSTANT ANYMORE. TO FIND A SOLUTION WE RLY ON THE EQUIVALENCE PRINCIPLE AND THE SCATTERED FIELD.

SCATTERED FIELD

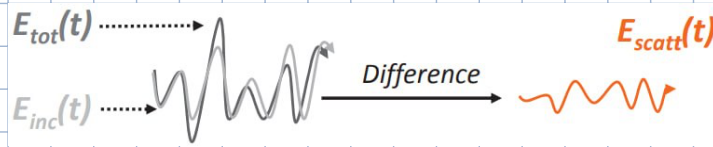
THE SCATTERED FIELD IS DEFINED AS THE DIFFERENCE BETWEEN THE TOTAL FIELD AND THE INCIDENT FIELD. DUE TO THIS DEFINITION IT IS NOT DIRECTLY MEASURABLE, IT MUST BE COMPUTED. IT REPRESENTS THE FIELD THAT THE OBJECT IS SCATTERING.

$$\underline{E}_{\text{scatt}}(\underline{r}, t) = \underline{E}_{\text{tot}}(\underline{r}, t) - \underline{E}_{\text{inc}}(\underline{r}, t)$$

IT IS A CLEAR INDICATOR OF THE PRESENCE OF AN OBJECT (NOT OF ABSENCE THOUGH):

$$\underline{E}_{\text{scatt}}(\underline{r}, t) \neq \emptyset \Rightarrow \text{PRESENCE (SUFFICIENT CONDITION)}$$

$$\underline{E}_{\text{scatt}}(\underline{r}, t) = \emptyset \nRightarrow \text{ABSENCE (NOT SUFFICIENT)}$$

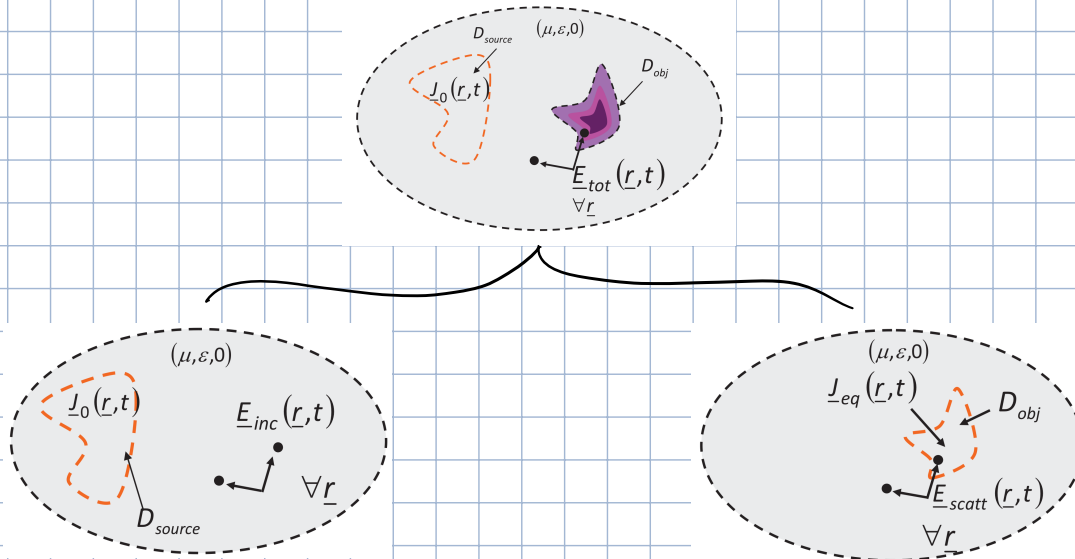


EQUIVALENCE PRINCIPLE

PREFACE NOTE: THERE ARE TWO MAIN FORMULATIONS OF THE EQUIVALENCE PRINCIPLE: SURFACE AND VOLUMETRIC.

- SURFACE FORMULATION WORKS FOR PERFECT ELECTRIC CONDUCTORS
- VOLUMETRIC FORMULATION WORKS FOR NON-PECS ← WE USE THIS

THE EQUIVALENCE PRINCIPLE STATES THAT THE PROBLEM OF A SOURCE RADIATING IN A SCENARIO WITH AN OBJECT (DISCONTINUITY IN THE PROPAGATION MEDIUM) CAN BE SPLIT IN TWO SUB-PROBLEMS:



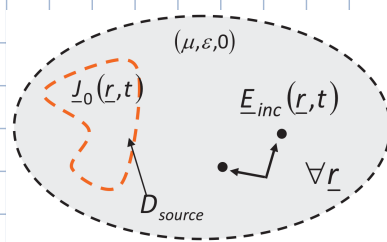
- 1) THE SOURCE $\tilde{\underline{J}}_0(\underline{r})$ RADIATES $\tilde{\underline{E}}_{\text{inc}}(\underline{r})$ IN FREE-SPACE (NO OBSTACLES)
- 2) AN EQUIVALENT SOURCE $\tilde{\underline{J}}_{\text{eq}}(\underline{r})$ DEFINED IN D_{obj} RADIATES $\tilde{\underline{E}}_{\text{scatt}}(\underline{r})$ IN FREE SPACE (REFERENCE SCENARIO)

NOTE THAT, UNDER THIS CONDITION, BOTH SOURCES RADIATE IN FREE SPACE!

MATHEMATICALLY, THIS CAN BE DERIVED AS FOLLOWS:

CONSIDER AN ELECTRIC CURRENT $\tilde{\underline{J}}_0$ AND A MAGNETIC CURRENT $\tilde{\underline{J}}_m$ TO BE THE IMPRESSED SOURCES RADIATING IN FREE SPACE $\{\epsilon, \mu, \sigma = 0\}$ (W/O OBJECT)

NEEDED ONLY TO HAVE A SYMMETRIC FORMULATION IN MAXWELL'S EQUATIONS



THE RADIATED FIELD

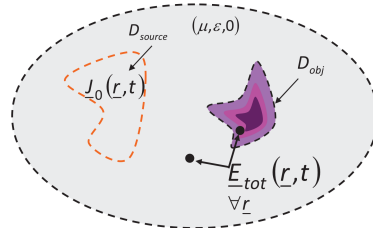
$$\begin{cases} \tilde{\underline{E}}(\underline{r}) = \tilde{\underline{E}}_{inc}(\underline{r}) \\ \tilde{\underline{H}}(\underline{r}) = \tilde{\underline{H}}_{inc}(\underline{r}) \end{cases}$$

SATISFIES MAXWELL'S EQUATIONS

$$\begin{cases} \nabla \times \tilde{\underline{E}}_{inc}(\underline{r}) = -j\omega\mu \tilde{\underline{H}}_{inc}(\underline{r}) - \tilde{\underline{M}}_0(\underline{r}) \\ \nabla \times \tilde{\underline{H}}_{inc}(\underline{r}) = j\omega\epsilon \tilde{\underline{E}}_{inc}(\underline{r}) + \tilde{\underline{J}}_0(\underline{r}) \end{cases}$$

↑
CONSTANTS

CONSIDER NOW $\tilde{\underline{J}}_0(\underline{r})$, $\tilde{\underline{M}}_0(\underline{r})$ RADIATING IN A SCENARIO WITH THE OBJECT (DISCONTINUITY IN MEDIUM) $\{\epsilon(\underline{r}), \mu(\underline{r}), \sigma \neq 0\}$.



USING PHASOR NOTATION, THE MEDIUM PARAMETERS BECOME $\{\tilde{\underline{E}}(\underline{r}), \tilde{\mu}(\underline{r})\}$, WHERE $\tilde{\underline{E}}(\underline{r}) = \underline{E}(\underline{r}) - j\frac{\sigma}{\omega}$

THE RADIATED FIELD

$$\begin{cases} \tilde{\underline{E}}(\underline{r}) = \tilde{\underline{E}}_{tot}(\underline{r}) \\ \tilde{\underline{H}}(\underline{r}) = \tilde{\underline{H}}_{tot}(\underline{r}) \end{cases}$$

SATISFIES MAXWELL'S EQUATIONS

$$\begin{cases} \nabla \times \tilde{\underline{E}}_{tot}(\underline{r}) = -j\omega\tilde{\mu}(\underline{r}) \tilde{\underline{H}}_{tot}(\underline{r}) - \tilde{\underline{M}}_0(\underline{r}) \\ \nabla \times \tilde{\underline{H}}_{tot}(\underline{r}) = j\omega\tilde{\epsilon}(\underline{r}) \tilde{\underline{E}}_{tot}(\underline{r}) + \tilde{\underline{J}}_0(\underline{r}) \end{cases}$$

↑
NOT CONSTANTS!

WE CAN NOW COMPUTE THE SCATTERED FIELD BY SUBTRACTING THE TWO FORMULATIONS

$$\begin{cases} \nabla \times \tilde{\underline{E}}_{inc}(\underline{r}) = -j\omega\mu \tilde{\underline{H}}_{inc}(\underline{r}) - \tilde{\underline{M}}_0(\underline{r}) \\ \nabla \times \tilde{\underline{H}}_{inc}(\underline{r}) = j\omega\epsilon \tilde{\underline{E}}_{inc}(\underline{r}) + \tilde{\underline{J}}_0(\underline{r}) \end{cases} \quad -$$

$$\begin{cases} \nabla \times \tilde{\underline{E}}_{tot}(\underline{r}) = -j\omega\tilde{\mu}(\underline{r}) \tilde{\underline{H}}_{tot}(\underline{r}) - \tilde{\underline{M}}_0(\underline{r}) \\ \nabla \times \tilde{\underline{H}}_{tot}(\underline{r}) = j\omega\tilde{\epsilon}(\underline{r}) \tilde{\underline{E}}_{tot}(\underline{r}) + \tilde{\underline{J}}_0(\underline{r}) \end{cases} \quad =$$

$$\begin{cases} \nabla \times \{\tilde{\underline{E}}_{tot}(\underline{r}) - \tilde{\underline{E}}_{inc}(\underline{r})\} = -j\omega [\tilde{\mu}(\underline{r}) \tilde{\underline{H}}_{tot}(\underline{r}) + \cancel{\tilde{\underline{M}}_0(\underline{r})} - \mu \tilde{\underline{H}}_{inc}(\underline{r}) - \cancel{\tilde{\underline{M}}_0(\underline{r})}] \\ \nabla \times \{\tilde{\underline{H}}_{tot}(\underline{r}) - \tilde{\underline{H}}_{inc}(\underline{r})\} = j\omega [\tilde{\epsilon}(\underline{r}) \tilde{\underline{E}}_{tot}(\underline{r}) + \cancel{\tilde{\underline{J}}_0(\underline{r})} - \epsilon \tilde{\underline{E}}_{inc}(\underline{r}) - \cancel{\tilde{\underline{J}}_0(\underline{r})}] \end{cases}$$

NOTE THAT $\tilde{\underline{\underline{J}}}_o$ AND $\tilde{\underline{\underline{M}}}_o$ CANCEL OUT

$$\begin{cases} \nabla \times \{ \tilde{\underline{\underline{E}}}_{\text{TOT}}(\underline{r}) - \tilde{\underline{\underline{E}}}_{\text{INC}}(\underline{r}) \} = -j\omega [\tilde{\underline{\underline{\mu}}}(\underline{r}) \tilde{\underline{\underline{H}}}_{\text{TOT}}(\underline{r}) - \underline{\mu} \tilde{\underline{\underline{H}}}_{\text{INC}}(\underline{r})] \\ \nabla \times \{ \tilde{\underline{\underline{H}}}_{\text{TOT}}(\underline{r}) - \tilde{\underline{\underline{H}}}_{\text{INC}}(\underline{r}) \} = j\omega [\tilde{\underline{\underline{\epsilon}}}(\underline{r}) \tilde{\underline{\underline{E}}}_{\text{TOT}}(\underline{r}) - \underline{\epsilon} \tilde{\underline{\underline{E}}}_{\text{INC}}(\underline{r})] \end{cases}$$

FROM THE DEFINITION OF THE SCATTERED FIELD WE KNOW THAT

$$\begin{cases} \tilde{\underline{\underline{E}}}_{\text{SCATT}}(\underline{r}) = \tilde{\underline{\underline{E}}}_{\text{TOT}}(\underline{r}) - \tilde{\underline{\underline{E}}}_{\text{INC}}(\underline{r}) \\ \tilde{\underline{\underline{H}}}_{\text{SCATT}}(\underline{r}) = \tilde{\underline{\underline{H}}}_{\text{TOT}}(\underline{r}) - \tilde{\underline{\underline{H}}}_{\text{INC}}(\underline{r}) \end{cases} \iff \begin{cases} \tilde{\underline{\underline{E}}}_{\text{INC}}(\underline{r}) = \tilde{\underline{\underline{E}}}_{\text{TOT}}(\underline{r}) - \tilde{\underline{\underline{E}}}_{\text{SCATT}}(\underline{r}) \\ \tilde{\underline{\underline{H}}}_{\text{INC}}(\underline{r}) = \tilde{\underline{\underline{H}}}_{\text{TOT}}(\underline{r}) - \tilde{\underline{\underline{H}}}_{\text{SCATT}}(\underline{r}) \end{cases}$$

AND BY SUBSTITUTING, WE GET

$$\begin{cases} \nabla \times \tilde{\underline{\underline{E}}}_{\text{SCATT}}(\underline{r}) = -j\omega \{ \tilde{\underline{\underline{\mu}}}(\underline{r}) \tilde{\underline{\underline{H}}}_{\text{TOT}}(\underline{r}) - \underline{\mu} [\tilde{\underline{\underline{H}}}_{\text{TOT}}(\underline{r}) - \tilde{\underline{\underline{H}}}_{\text{SCATT}}(\underline{r})] \} \\ \nabla \times \tilde{\underline{\underline{H}}}_{\text{SCATT}}(\underline{r}) = j\omega \{ \tilde{\underline{\underline{\epsilon}}}(\underline{r}) \tilde{\underline{\underline{E}}}_{\text{TOT}}(\underline{r}) - \underline{\epsilon} [\tilde{\underline{\underline{E}}}_{\text{TOT}}(\underline{r}) - \tilde{\underline{\underline{E}}}_{\text{SCATT}}(\underline{r})] \} \end{cases}$$

$$\begin{cases} \nabla \times \tilde{\underline{\underline{E}}}_{\text{SCATT}}(\underline{r}) = -j\omega \{ [\tilde{\underline{\underline{\mu}}}(\underline{r}) - \underline{\mu}] \tilde{\underline{\underline{H}}}_{\text{TOT}}(\underline{r}) + \underline{\mu} \tilde{\underline{\underline{H}}}_{\text{SCATT}}(\underline{r}) \} \\ \nabla \times \tilde{\underline{\underline{H}}}_{\text{SCATT}}(\underline{r}) = j\omega \{ [\tilde{\underline{\underline{\epsilon}}}(\underline{r}) - \underline{\epsilon}] \tilde{\underline{\underline{E}}}_{\text{TOT}}(\underline{r}) + \underline{\epsilon} \tilde{\underline{\underline{E}}}_{\text{SCATT}}(\underline{r}) \} \end{cases}$$

$$\begin{cases} \nabla \times \tilde{\underline{\underline{E}}}_{\text{SCATT}}(\underline{r}) = -j\omega [\tilde{\underline{\underline{\mu}}}(\underline{r}) - \underline{\mu}] \tilde{\underline{\underline{H}}}_{\text{TOT}}(\underline{r}) - j\omega \underline{\mu} \tilde{\underline{\underline{H}}}_{\text{SCATT}}(\underline{r}) \\ \nabla \times \tilde{\underline{\underline{H}}}_{\text{SCATT}}(\underline{r}) = j\omega [\tilde{\underline{\underline{\epsilon}}}(\underline{r}) - \underline{\epsilon}] \tilde{\underline{\underline{E}}}_{\text{TOT}}(\underline{r}) + j\omega \underline{\epsilon} \tilde{\underline{\underline{E}}}_{\text{SCATT}}(\underline{r}) \end{cases}$$

WE CAN NOTICE A SIMILARITY WITH MAXWELL'S EQUATIONS

$$\begin{cases} \nabla \times \tilde{\underline{\underline{E}}}_{\text{SCATT}}(\underline{r}) = -j\omega \underline{\mu} \tilde{\underline{\underline{H}}}_{\text{SCATT}}(\underline{r}) - j\omega [\tilde{\underline{\underline{\mu}}}(\underline{r}) - \underline{\mu}] \tilde{\underline{\underline{H}}}_{\text{TOT}}(\underline{r}) \\ \nabla \times \tilde{\underline{\underline{H}}}_{\text{SCATT}}(\underline{r}) = j\omega \underline{\epsilon} \tilde{\underline{\underline{E}}}_{\text{SCATT}}(\underline{r}) + j\omega [\tilde{\underline{\underline{\epsilon}}}(\underline{r}) - \underline{\epsilon}] \tilde{\underline{\underline{E}}}_{\text{TOT}}(\underline{r}) \end{cases}$$

$$\begin{cases} \nabla \times \tilde{\underline{\underline{E}}}_{\text{INC}}(\underline{r}) = -j\omega \underline{\mu} \tilde{\underline{\underline{H}}}_{\text{INC}}(\underline{r}) - \tilde{\underline{\underline{M}}}_o(\underline{r}) \\ \nabla \times \tilde{\underline{\underline{H}}}_{\text{INC}}(\underline{r}) = j\omega \underline{\epsilon} \tilde{\underline{\underline{E}}}_{\text{INC}}(\underline{r}) + \tilde{\underline{\underline{J}}}_o(\underline{r}) \end{cases}$$

WE CAN IN FACT DEFINE VOLUMETRIC EQUIVALENT CURRENTS

$$\begin{cases} \tilde{\underline{\underline{M}}}_{\text{EQ}}(\underline{r}) := j\omega [\tilde{\underline{\underline{\mu}}}(\underline{r}) - \underline{\mu}] \tilde{\underline{\underline{H}}}_{\text{TOT}}(\underline{r}) & \left[\frac{\text{V}}{\text{m}^2} \right] \\ \tilde{\underline{\underline{J}}}_{\text{EQ}}(\underline{r}) := j\omega [\tilde{\underline{\underline{\epsilon}}}(\underline{r}) - \underline{\epsilon}] \tilde{\underline{\underline{E}}}_{\text{TOT}}(\underline{r}) & \left[\frac{\text{A}}{\text{m}^2} \right] \end{cases}$$

UNKNOWN!

NOTE THAT THESE ARE DEFINED (NON-ZERO) ONLY INSIDE THE OBJECT: $\tilde{\underline{\underline{E}}}(\underline{r}) = \underline{E} \quad \forall \underline{r} \notin \text{D}_{\text{OBJ}} \Rightarrow \underline{E} - \underline{E} = \underline{0}$

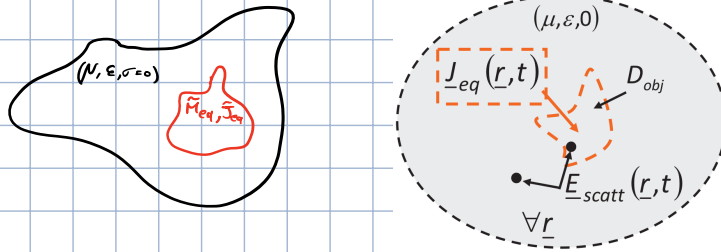
SINCE THEY DEPEND ON THE OBJECT'S CHARACTERISTICS, THEY ARE UNKNOWN!

ALSO NOTE THAT, SINCE WE ARE CONSIDERING A LOSSLESS MEDIUM ($\sigma = 0$), $\tilde{\underline{\underline{J}}}_e(\underline{r}) = \underline{0}$

BY SUBSTITUTING THE NEWLY DEFINED CURRENTS, WE OBTAIN

$$\begin{cases} \nabla \times \tilde{\underline{E}}_{\text{scatt}}(\underline{r}) = -j\omega\mu \tilde{\underline{H}}_{\text{scatt}}(\underline{r}) - \tilde{\underline{M}}_{\text{eq}}(\underline{r}) \\ \nabla \times \tilde{\underline{H}}_{\text{scatt}}(\underline{r}) = j\omega\epsilon \tilde{\underline{E}}_{\text{scatt}}(\underline{r}) + \tilde{\underline{J}}_{\text{eq}}(\underline{r}) \end{cases}$$

WHICH HAS THE SAME EXACT FORM OF THE EQUATIONS WE STARTED WITH. THIS MEANS THAT WE CAN REFORMULATE THE PROBLEM AS AN EQUIVALENT SOURCE, DEFINED INSIDE THE OBJECT, RADIATING IN FREE-SPACE.

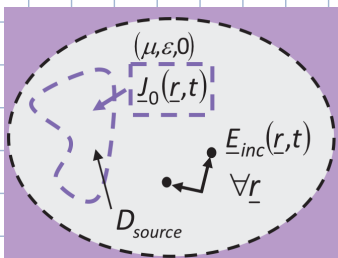


THIS CONCLUSION IS CRUCIAL. SINCE THE EQUIVALENT SOURCE IS RADIATING IN FREE-SPACE, THE MEDIUM IS LINEAR. THIS MEANS THAT SUPERPOSITION HOLDS AND WE CAN USE THE USUAL CONVOLUTION $\underline{E} = \underline{J} * \underline{G}$

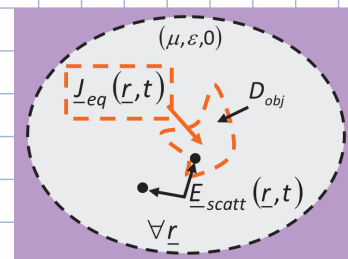
TO CONCLUDE, ALL FIELDS HAVE NOW THE SAME STRUCTURE

$$\begin{aligned} \tilde{\underline{J}}(\underline{r}), \tilde{\underline{M}}(\underline{r}) &\rightarrow \tilde{\underline{E}}(\underline{r}), \tilde{\underline{H}}(\underline{r}) \\ &\downarrow \\ \begin{cases} \nabla \times \tilde{\underline{E}}(\underline{r}) = -j\omega\mu \tilde{\underline{H}}(\underline{r}) - \tilde{\underline{M}}(\underline{r}) \\ \nabla \times \tilde{\underline{H}}(\underline{r}) = j\omega\epsilon \tilde{\underline{E}}(\underline{r}) - \tilde{\underline{J}}(\underline{r}) \end{cases} \end{aligned}$$

$$\begin{aligned} \tilde{\underline{J}}_0(\underline{r}), \tilde{\underline{M}}_0(\underline{r}) &\rightarrow \tilde{\underline{E}}_{\text{inc}}(\underline{r}), \tilde{\underline{H}}_{\text{inc}}(\underline{r}) \\ &\downarrow \\ \begin{cases} \nabla \times \tilde{\underline{E}}_{\text{inc}}(\underline{r}) = -j\omega\mu \tilde{\underline{H}}_{\text{inc}}(\underline{r}) - \tilde{\underline{M}}_0(\underline{r}) \\ \nabla \times \tilde{\underline{H}}_{\text{inc}}(\underline{r}) = j\omega\epsilon \tilde{\underline{E}}_{\text{inc}}(\underline{r}) - \tilde{\underline{J}}_0(\underline{r}) \end{cases} \end{aligned}$$



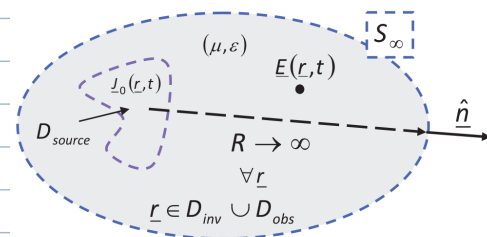
$$\begin{aligned} \tilde{\underline{J}}_{\text{eq}}(\underline{r}), \tilde{\underline{M}}_{\text{eq}}(\underline{r}) &\rightarrow \tilde{\underline{E}}_{\text{scatt}}(\underline{r}), \tilde{\underline{H}}_{\text{scatt}}(\underline{r}) \\ &\downarrow \\ \begin{cases} \nabla \times \tilde{\underline{E}}_{\text{scatt}}(\underline{r}) = -j\omega\mu \tilde{\underline{H}}_{\text{scatt}}(\underline{r}) - \tilde{\underline{M}}_{\text{eq}}(\underline{r}) \\ \nabla \times \tilde{\underline{H}}_{\text{scatt}}(\underline{r}) = j\omega\epsilon \tilde{\underline{E}}_{\text{scatt}}(\underline{r}) - \tilde{\underline{J}}_{\text{eq}}(\underline{r}) \end{cases} \end{aligned}$$



SCATTERED FIELD DESCRIPTIVE EQUATIONS

WE ALREADY KNOW THAT THE FIELD RADIATED IN FREE SPACE BY A GENERIC SOURCE $\tilde{\underline{J}}(\underline{r})$ SATISFIES THE FOLLOWING EQUATIONS:

$$\begin{cases} \nabla \times \nabla \times \tilde{\underline{E}}(\underline{r}) + k^2 \tilde{\underline{E}}(\underline{r}) = -j\omega\mu \tilde{\underline{J}}(\underline{r}) \\ \begin{cases} \tilde{\underline{E}}(\underline{r}) \times \hat{\underline{n}}(\underline{r}) = \underline{E}(\underline{r}) \cdot \hat{\underline{n}}(\underline{r}) \hat{\underline{n}} \\ \tilde{\underline{E}}(\underline{r}) = \eta \tilde{\underline{H}}(\underline{r}) \times \hat{\underline{n}} \end{cases} \end{cases}, \underline{r} \in S_\infty$$



WHICH HAS THE FOLLOWING SOLUTION

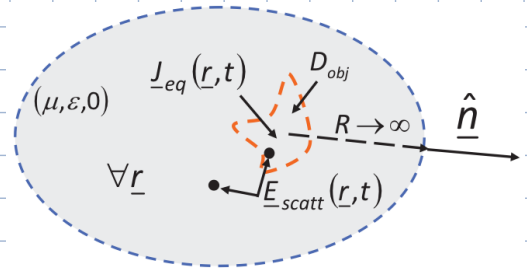
$$\tilde{\underline{E}}(\underline{r}) = j\omega\mu \int \tilde{\underline{J}}(\underline{r}') \cdot \underline{G}(\underline{r}, \underline{r}') d\underline{r}'$$

D_{SOURCE}

BY ANALOGY WE CAN USE THE EQUIVALENT SOURCE TO OBTAIN A SOLUTION FOR THE SCATTERED FIELD

$$\tilde{\underline{J}}(\underline{r}) := \tilde{\underline{J}}_{EQ}(\underline{r}) \quad \tilde{\underline{E}}(\underline{r}) := \tilde{\underline{E}}_{SCATT}(\underline{r})$$

$$\begin{cases} \nabla \times \nabla \times \tilde{\underline{E}}_{SCATT}(\underline{r}) + k^2 \tilde{\underline{E}}_{SCATT}(\underline{r}) = -j\omega\mu \tilde{\underline{J}}_{EQ}(\underline{r}) \\ \tilde{\underline{E}}_{SCATT}(\underline{r}) \times \underline{\hat{n}}_{SCATT}(\underline{r}) = \underline{E}_{SCATT}(\underline{r}) \underline{H}_{SCATT}(\underline{r}) \underline{\hat{n}} \\ \tilde{\underline{E}}_{SCATT}(\underline{r}) = \eta \underline{\hat{H}}(\underline{r}) \times \underline{\hat{n}} \end{cases}, \underline{r} \in S_{OBJ}$$

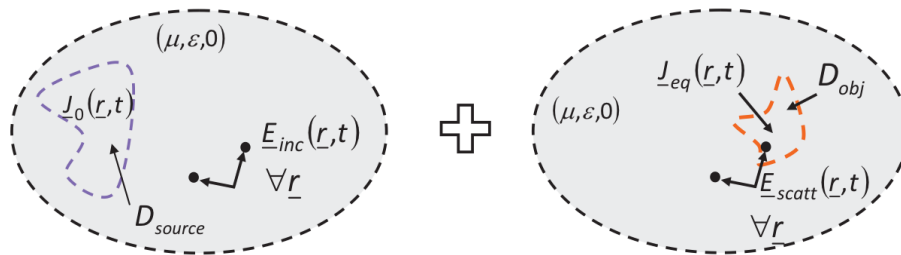


WHICH, AS A RESULT, HAS THE FOLLOWING SOLUTION:

$$\tilde{\underline{E}}_{SCATT}(\underline{r}) = j\omega\mu \int_{D_{OBJ}} \tilde{\underline{J}}_{EQ} \cdot \underline{G}(\underline{r}|\underline{r}') d\underline{r}'$$

TOTAL FIELD DESCRIBING EQUATIONS

WITH THIS DEFINITION, WE CAN NOW DETERMINE THE TOTAL FIELD AS THE SUM OF THE INCIDENT AND SCATTERED FIELDS.



$$\tilde{\underline{E}}_{TOT}(\underline{r}) = \tilde{\underline{E}}_{INC}(\underline{r}) + \tilde{\underline{E}}_{SCATT}(\underline{r}) = j\omega\mu \int_{D_{SOURCE}} \tilde{\underline{J}}_0 \cdot \underline{G}(\underline{r}|\underline{r}') d\underline{r}' + j\omega\mu \int_{D_{OBJ}} \tilde{\underline{J}}_{EQ} \cdot \underline{G}(\underline{r}|\underline{r}') d\underline{r}'$$

UNKNOWN!

WHERE $\tilde{\underline{J}}_{EQ}(\underline{r}) := j\omega[\tilde{\underline{E}}(\underline{r}) - \underline{E}] \tilde{\underline{E}}_{TOT}(\underline{r})$ AND $\tilde{\underline{E}}(\underline{r}) = \underline{E}(\underline{r}) - j\frac{\underline{\sigma}}{\omega}$

NOTE THAT WE CAN SAY THAT THE SOLUTION, IN TERMS OF FIELDS, IS THE SUMMATION OF THE SOLUTION OF THE TWO SUB-PROBLEMS, BECAUSE WE ARE DEALING WITH CURRENTS AND THE PROBLEM OF THE FIELD IS LINEAR W.R.T. THE CURRENT, WHILE IT IS NON-LINEAR W.R.T. THE PERMEABILITY AND PERMITTIVITY OF THE OBJECT.

$$\text{ORIGINAL: } \underline{E}_{TOT} = (\underline{J}_{eq} + \underline{J}_0) \cdot \underline{G} \quad \longrightarrow \quad \text{EQUIVALENT: } \begin{cases} \underline{E}_{SCATT} = \underline{J}_{eq} \cdot \underline{G} \\ \underline{E}_{INC} = \underline{J}_0 \cdot \underline{G} \end{cases}$$

NOW WE KNOW HOW TO FORMULATE THE TOTAL FIELD, HOWEVER WE STILL DON'T KNOW $\underline{E}, \underline{J}$ INSIDE THE OBJECT.

TO FIND OUT, WE NEED TO SOLVE AN INVERSE PROBLEM: COMPUTE $\tilde{\underline{E}}(\underline{r}), \tilde{\underline{J}}(\underline{r})$ GIVEN MEASUREMENTS OF THE TOTAL AND INCIDENT FIELDS.

OBJECT FUNCTION

WE NOW KNOW EXPLICITLY THE RELATION BETWEEN THE OBJECT AND THE TOTAL FIELD. LET'S NOW START TO DESCRIBE THE OBJECT ITSELF.

THE OBJECT FUNCTION ALLOWS US TO DESCRIBE THE OBJECT IN TERMS OF DIFFERENCE W.R.T. THE MEDIUM.

IT IS A DISCONTINUITY MEASURE. AS USUAL, WE ARE DOING A DIFFERENTIAL ANALYSIS.

THE GOAL OF THE INVERSE PROBLEM WILL THEN BE TO RETRIEVE THIS VERY FUNCTION.

THIS IS HOW WE DEFINED THE CONCEPT OF OBJECT IN THE BEGINNING

$$\tau(\underline{r}) = \overbrace{\tilde{E}(\underline{r}) - E}^{\text{DIFFERENCE}} = E(\underline{r}) - j \frac{\sigma}{\omega} - E = (E(\underline{r}) - E) - j \frac{\sigma}{\omega}$$

WE CAN SEE THAT $\tau(\underline{r})$, ASSUMING IT CAN BE COMPUTED, IS A COMPLEX SCALAR FUNCTION, WHERE:

$$-\text{Re}\{\tau(\underline{r})\} = \tilde{E}(\underline{r}) - E \quad \text{REPRESENTS THE PERMITTIVITY}$$

$$-\text{Im}\{\tau(\underline{r})\} = -\frac{\sigma}{\omega} \quad \text{REPRESENTS THE CONDUCTIVITY}$$

THE OBJECT FUNCTION IS A GREAT INDICATOR OF THE PRESENCE OF AN OBJECT, BECAUSE

$$\left. \begin{array}{l} \tau(\underline{r}) \neq 0 \Rightarrow \text{DISCONTINUITY IN MEDIUM} \Rightarrow \text{OBJECT} \quad E(\underline{r}) \neq E \\ \tau(\underline{r}) = 0 \Rightarrow \text{NO DISCONTINUITY IN MEDIUM} \Rightarrow \text{NO OBJECT} \quad E(\underline{r}) = E \end{array} \right\} \text{THIS IS OUR USUAL DEFINITION OF THE EXISTENCE OF AN OBJECT FROM THE SYSTEM'S POV}$$

↓
SUFFICIENT CONDITION!

EQUIVALENT CURRENT DENSITY (VOLUMETRIC)

WE HAVE ALREADY INTRODUCED $\tilde{\underline{J}}_{eq}(\underline{r})$ AND WE KNOW IT IS A FICTITIOUS SOURCE THAT MODELS THE OBJECT AS A SOURCE THAT RADIATES THE SCATTERED FIELD IN FREE-SPACE.

$$\tilde{\underline{E}}_{scat}(\underline{r}) = j\omega\mu \int_{D_{OBJ}} \tilde{\underline{J}}_{eq} \cdot \underline{G}(\underline{r}|\underline{r}') d\underline{r}'$$

IT IS DEFINED THROUGH THE CONTRAST OF THE DIELECTRIC PERMITTIVITY, WHICH WE JUST REDEFINED AS THE OBJECT FUNCTION

$$\tilde{\underline{J}}_{eq}(\underline{r}) = j\omega[\tilde{E}(\underline{r}) - E]\tilde{\underline{E}}_{tot}(\underline{r}) = j\omega\tau(\underline{r})\tilde{\underline{E}}_{tot}(\underline{r}), \quad \underline{r} \in D_{OBJ}$$

NOTE THAT $\tilde{\underline{J}}_{eq}(\underline{r})$ IS A COMPLEX VECTORIAL FUNCTION.

$\tilde{\underline{J}}_{eq}(\underline{r})$ IS AN INDICATOR OF THE EXISTENCE OF AN OBJECT, BUT NOT OF ITS NON-EXISTENCE

$$\tilde{\underline{J}}_{eq}(\underline{r}) \neq 0 \Rightarrow \text{OBJECT} \quad \tau(\underline{r}) \neq 0 \cup \tilde{\underline{E}}_{tot}(\underline{r}) \neq 0$$

$$\tilde{\underline{J}}_{eq}(\underline{r}) = 0 \not\Rightarrow \text{NO OBJECT} \quad \tau(\underline{r}) \neq 0 \cup \tilde{\underline{E}}_{tot}(\underline{r}) = 0 \quad \text{OR} \quad \tau(\underline{r}) = 0 \cup \tilde{\underline{E}}_{tot}(\underline{r}) \neq 0$$

IN SUMMARY, IT IS NOT AN INDICATOR FUNCTION

$$\tau(\underline{r}) = 0 \not\Rightarrow \tilde{\underline{J}}_{eq}(\underline{r}) = 0$$